Linear Algebra I & Mathematics Tutorial Ib
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Quiz 4: Linear Maps

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1 Comment 1

When you need to find image of any vector $v \in V$ (vector space) under a linear map $f : V \to W$ where (W is another vector space), you only need to find images of basis elements of V under f.

For example, let $\{v_1, v_2 \cdots, v_n\}$ be the basis of V. Then take any $v \in V$. We can write $v = \sum_{i=1}^{n} c_i v_i$ for $c_i \in \mathbb{F}$. And hence,

$$f(v) = f\left(\sum_{i=1}^{n} c_i v_i\right) = \sum_{i=1}^{n} c_i f(v_i) \quad \text{(Using the linearity of } f\text{)} \tag{1}$$

So, if you know $f(v_i) \quad \forall i = 1, \cdots, n$, then you know $f(v)$ from (1).

Example 1.1. In exercise 2. in this quiz, if you can notice that the basis of $\mathbb{R}^2$ is $\{(1,0),(0,1)\}$. So, if you find $f((1,0))$ and $f((0,1))$ you can find $f(v)$ for any $v \in \mathbb{R}^2$.

Students' Solutions

But how do you find $f(1,0)$ and $f(0,1)$ when it is not obvious? I like how **Raphael Njauw** solved. Please check his solution [RN] and **Bipasana Bhattacharya** found the function F in a different way from Comment 1. But her approach (see [BB]) is a generalisation of the approach of [RN]. **Dizon Alfonso Miguel Cruz** also found the matrix in a unique way. See his solution here [DAMC].

For example, $f((3,1)) = f(3(1,0)+(0,1)) = 3f((1,0))+f((0,1)) = (5,6,-2)$.

2 Comment 2

There is a systematic way to find a matrix of a linear map between **finite dimensional** vector spaces.

Suppose $f : V \to W$ be a linear map. V has basis $\mathcal{B}_V = \{v_1, \cdots, v_n\}$ and W has basis $\mathcal{B}_W = \{w_1, \cdots w_m\}$ then to find the matrix of $f := A[f]_{\mathcal{B}_V}^{\mathcal{B}_W}$, write,

$$f(v_i) = \sum_{j=1}^{m} c_{ij} w_j \tag{2}$$

And $A[f]_{\mathcal{B}_V}^{\mathcal{B}_W} = [c_{ij}]_{m \times n}$. Explicitely, your ith column of the matrix $A[f]_{\mathcal{B}_V}^{\mathcal{B}_W}$ will be $(c_{i1} \quad c_{i2} \quad \cdots \quad c_{im})^T$.

Example 2.1. Here in the quiz in exercise 1, the basis of $\mathbb{R}^3$ is $\{(1,0,0),(0,1,0),(0,0,1)\}$ and similarly for $\mathbb{R}^2$ is $\{(1,0),(0,1)\}$. [We took standard basis here as, if no special basis is specified we work with standard basis.]

$$\begin{aligned} f(1,0,0) &= (1,0) = 1 \cdot (1,0) + 0 \cdot (0,1) \\ f(0,1,0) &= (0,3) = 0 \cdot (1,0) + 3 \cdot (0,1) \\ f(0,0,1) &= (-2,1) = -2 \cdot (1,0) + 1 \cdot (0,1) \end{aligned} \tag{3}$$

From (3) you get your matrix:

$$A[f]^{\mathcal{B}_{\mathbb{R}^2}}_{\mathcal{B}_{\mathbb{R}^3}} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 1 \end{pmatrix}$$

Note: This is a systematic way of writing matrix of a linear function.

3 Comment 3

Yuka Nakao solved the Exercise 1's second part in a unique way. She assumed that whatif g be a linear function and found its matrix representation from the given fromula of g. So, she used the idea in Comment 1 to find the matrix of g which is:

$$A[g]^{\mathcal{B}_{\mathbb{R}^2}}_{\mathcal{B}_{\mathbb{R}^2}} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

where no basis is mentioned so we took standard basis of $\mathbb{R}^2$ which is: $\mathcal{B}_{\mathbb{R}^2} = \{(1,0),(0,1)\}$.

But then she showed that

$$A[g]^{\mathcal{B}_{\mathbb{R}^2}}_{\mathcal{B}_{\mathbb{R}^2}} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \neq \begin{pmatrix} x_1 + 2x_2 \\ x_2^2 \end{pmatrix}$$

So, g has no matrix representation and this means g is not linear. Her original proof is a little messy so, I refined it here.

References

[RN] Raphael Njauw, *Solution to Quiz 4, Exercise 2*, PDF.

[BB] Bipasana Bhattacharya, *Solution to Quiz 4, Exercise 2*, PDF.

[DAMC] Dizon Alfonso Miguel Cruz, *Solution to Quiz 4, Exercise 2*, PDF.